📋 EduTrack — Software Requirements Specification (SRS)

Academic Management Platform

VTU Internship 2026 — Team 15

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the EduTrack Academic Management Platform. It is intended to guide the development team and serve as a reference for testing and validation.

1.2 Scope

EduTrack is a web-based academic management platform that provides role-based access for three types of users — Admin, Teacher, and Student. The system manages attendance, marks, assignments, course materials, and platform settings through a centralized interface.

1.3 Definitions

TERM	DEFINITION
Admin	System administrator who manages the platform
Teacher	Faculty member who manages courses and students
Student	Learner who accesses academic data
OTP	One-Time Password used for password reset
BCrypt	Password hashing algorithm used for secure storage
JWT	JSON Web Token (not used — custom auth implemented)
CRUD	Create, Read, Update, Delete operations
CSV	Comma-Separated Values file format for bulk import

TERM	DEFINITION
SPA	Single Page Application

1.4 References

- Spring Boot 3.2.3 Documentation
- React 19 Documentation
- MySQL 8.0 Reference Manual
- IEEE SRS Standard 830-1998

2. Overall Description

2.1 Product Perspective

EduTrack is a standalone web application consisting of three tiers:

- **Presentation Layer:** React 19 frontend
- **Business Logic Layer:** Spring Boot 3 REST API
- **Data Layer:** MySQL 8 database

2.2 Product Functions

- Role-based authentication and authorization
- Student and teacher management by admin
- Course creation and management by teachers
- Attendance marking and tracking
- Marks entry with automatic grade calculation
- Assignment creation and submission
- Course material upload and download
- OTP-based password reset via email
- Bulk data import via CSV
- Platform-wide system settings management

2.3 User Classes

USER CLASS	DESCRIPTION	ACCESS LEVEL
Admin	Manages the entire platform	Full access to all management features

USER CLASS	DESCRIPTION	ACCESS LEVEL
Teacher	Manages courses and students	Access to their own courses and students
Student	Views academic data	Read-only access to their own data

2.4 Operating Environment

- **Client:** Any modern web browser (Chrome, Firefox, Safari, Edge)
- **Server:** Java 21, Spring Boot 3.2.3
- **Database:** MySQL 8.0
- **OS:** Any OS supporting Java 21

2.5 Assumptions and Dependencies

- Users have a stable internet connection
- Gmail SMTP is available for OTP delivery
- MySQL 8+ is installed and running
- Java 21 is installed on the server

3. Functional Requirements

3.1 Authentication Module

- **FR-AUTH-01:** The system shall allow Admin to log in using email and password.
- **FR-AUTH-02:** The system shall allow Teacher to log in using email and password.
- **FR-AUTH-03:** The system shall allow Student to log in using email and password.
- **FR-AUTH-04:** The system shall display an error message for invalid credentials.
- **FR-AUTH-05:** The system shall block Student and Teacher login when maintenance mode is enabled.
- **FR-AUTH-06:** The system shall allow all users to log out.
- **FR-AUTH-07:** The system shall allow Student self-registration when registration is open.
- **FR-AUTH-08:** The system shall hash all passwords using BCrypt before storing.
- **FR-AUTH-09:** The system shall redirect users to their role-specific portal after login.
- **FR-AUTH-10:** The system shall clear credentials when switching roles on the login page.

3.2 OTP Password Reset Module

- **FR-OTP-01:** The system shall allow users to request a password reset via email.
- **FR-OTP-02:** The system shall validate that the email exists for the selected role before sending OTP.
- **FR-OTP-03:** The system shall generate a 6-digit OTP using a secure random generator.
- **FR-OTP-04:** The system shall send the OTP to the user's registered email via Gmail SMTP.
- **FR-OTP-05:** The OTP shall expire after 10 minutes.
- **FR-OTP-06:** The OTP shall be single-use — invalidated after successful verification.
- **FR-OTP-07:** The system shall delete old OTPs before generating a new one for the same email.
- **FR-OTP-08:** The system shall allow password reset after successful OTP verification.

3.3 Admin Module

- **FR-ADMIN-01:** The system shall allow Admin to view all registered students.
- **FR-ADMIN-02:** The system shall allow Admin to add a new student.
- **FR-ADMIN-03:** The system shall allow Admin to edit a student's name and phone number.
- **FR-ADMIN-04:** The system shall allow Admin to delete a student and all their related data.
- **FR-ADMIN-05:** The system shall allow Admin to search students by name, email, or department.
- **FR-ADMIN-06:** The system shall allow Admin to update a student's status (Active/Suspended/Graduated).
- **FR-ADMIN-07:** The system shall allow Admin to force reset a student's password.
- **FR-ADMIN-08:** The system shall allow Admin to bulk import students via CSV.
- **FR-ADMIN-09:** The system shall allow Admin to view all registered teachers.
- **FR-ADMIN-10:** The system shall allow Admin to add a new teacher.
- **FR-ADMIN-11:** The system shall allow Admin to edit a teacher's name, phone, and employee ID.
- **FR-ADMIN-12:** The system shall allow Admin to delete a teacher and all their related data.
- **FR-ADMIN-13:** The system shall allow Admin to update a teacher's status (Active/Suspended).

- **FR-ADMIN-14:** The system shall allow Admin to force reset a teacher's password.
- **FR-ADMIN-15:** The system shall allow Admin to bulk import teachers via CSV.
- **FR-ADMIN-16:** The system shall allow Admin to enable or disable maintenance mode.
- **FR-ADMIN-17:** The system shall allow Admin to open or close student self-registration.
- **FR-ADMIN-18:** The system shall allow Admin to set the active academic year and semester.
- **FR-ADMIN-19:** The system shall not allow Admin to change a student's department, year, or semester after registration.

3.4 Teacher Module

- **FR-TEACHER-01:** The system shall allow Teacher to view all courses assigned to them.
- **FR-TEACHER-02:** The system shall allow Teacher to create a new course.
- **FR-TEACHER-03:** The system shall allow Teacher to update course details.
- **FR-TEACHER-04:** The system shall allow Teacher to delete a course and all its related data.
- **FR-TEACHER-05:** The system shall allow Teacher to mark attendance for students per course per date.
- **FR-TEACHER-06:** The system shall update existing attendance if the same student+course+date record exists (upsert).
- **FR-TEACHER-07:** The system shall allow Teacher to view attendance by course and by date.
- **FR-TEACHER-08:** The system shall allow Teacher to view attendance statistics per student.
- **FR-TEACHER-09:** The system shall allow Teacher to enter marks for students per course.
- **FR-TEACHER-10:** The system shall auto-calculate grades based on marks entered.
- **FR-TEACHER-11:** The system shall allow Teacher to update and delete marks.
- **FR-TEACHER-12:** The system shall allow Teacher to create assignments with title, description, and due date.
- **FR-TEACHER-13:** The system shall allow Teacher to update and delete assignments.
- **FR-TEACHER-14:** The system shall allow Teacher to view student submissions per assignment.

- **FR-TEACHER-15:** The system shall allow Teacher to grade submissions and add feedback.
- **FR-TEACHER-16:** The system shall allow Teacher to upload course materials as files.
- **FR-TEACHER-17:** The system shall allow Teacher to add external links as course materials.
- **FR-TEACHER-18:** The system shall allow Teacher to delete course materials.
- **FR-TEACHER-19:** The system shall allow Teacher to bulk import attendance via CSV.
- **FR-TEACHER-20:** The system shall allow Teacher to bulk import marks via CSV.
- **FR-TEACHER-21:** The system shall allow Teacher to update their name and phone number.
- **FR-TEACHER-22:** The system shall allow Teacher to change their password.

3.5 Student Module

- **FR-STUDENT-01:** The system shall allow Student to register with their details.
- **FR-STUDENT-02:** The system shall allow Student to view all courses in their department.
- **FR-STUDENT-03:** The system shall allow Student to view their attendance per course.
- **FR-STUDENT-04:** The system shall display attendance percentage per course for the student.
- **FR-STUDENT-05:** The system shall allow Student to view their marks per course.
- **FR-STUDENT-06:** The system shall display the auto-calculated grade for each course.
- **FR-STUDENT-07:** The system shall display the student's CGPA on the dashboard.
- **FR-STUDENT-08:** The system shall allow Student to view all assignments for their courses.
- **FR-STUDENT-09:** The system shall allow Student to submit assignments as file uploads.
- **FR-STUDENT-10:** The system shall allow Student to check their submission status and grade.
- **FR-STUDENT-11:** The system shall allow Student to view and download course materials.
- **FR-STUDENT-12:** The system shall allow Student to update their name and phone number.
- **FR-STUDENT-13:** The system shall allow Student to change their password.

- **FR-STUDENT-14:** The system shall not allow Student to change their department, year, or semester.

3.6 Grading System

FR-GRADE-01: The system shall automatically assign grades based on the following scale:

MARKS	GRADE
90 – 100	O
80 – 89	A+
70 – 79	A
60 – 69	B+
Below 60	B

4. Non-Functional Requirements

4.1 Performance

- **NFR-PERF-01:** The system shall load the dashboard within 3 seconds on a standard internet connection.
- **NFR-PERF-02:** API responses shall be returned within 2 seconds for standard operations.
- **NFR-PERF-03:** The system shall support at least 50 concurrent users.

4.2 Security

- **NFR-SEC-01:** All passwords shall be hashed using BCrypt before storage.
- **NFR-SEC-02:** Passwords shall never be returned in API responses.
- **NFR-SEC-03:** OTPs shall expire after 10 minutes and be single-use.
- **NFR-SEC-04:** CSRF protection shall be disabled for stateless REST API.
- **NFR-SEC-05:** CORS shall be configured to allow requests from the frontend domain.

4.3 Usability

- **NFR-USE-01:** The system shall provide a dark and light theme option.
- **NFR-USE-02:** The system shall display clear error messages for all failed operations.

- **NFR-USE-03:** The system shall work on all modern browsers (Chrome, Firefox, Safari, Edge).
- **NFR-USE-04:** The login page shall clear fields when switching between roles.

4.4 Reliability

- **NFR-REL-01:** The system shall maintain data integrity through foreign key constraints.
- **NFR-REL-02:** Cascade delete shall ensure no orphan records remain after deletion.
- **NFR-REL-03:** The default admin account shall be auto-created on first startup.

4.5 Maintainability

- **NFR-MAIN-01:** The codebase shall follow a layered architecture (Controller, Service, Repository, Entity).
- **NFR-MAIN-02:** All sensitive configuration shall be stored in `application.properties` (gitignored).
- **NFR-MAIN-03:** A template file `application.properties.template` shall be provided for setup.

4.6 Portability

- **NFR-PORT-01:** The backend shall be containerized using Docker for platform-independent deployment.
- **NFR-PORT-02:** The frontend shall be deployable on any static hosting platform.

5. System Constraints

CONSTRAINT	DESCRIPTION
Technology	React, Spring Boot, MySQL only
Authentication	Custom BCrypt-based auth (no JWT)
Deployment	Free tier hosting
Email	Gmail SMTP only
Departments	Only CSE and AIML supported

6. External Interface Requirements

6.1 User Interface

- Web browser-based interface
- Responsive layout
- Dark/Light theme support
- Role-based navigation sidebar

6.2 Hardware Interface

- Standard computer or laptop with internet access
- No special hardware required

6.3 Software Interface

- MySQL 8.0 database
- Gmail SMTP for email delivery
- Browser supporting ES6+ JavaScript

6.4 Communication Interface

- HTTPS for all API communication in production
- REST API with JSON request/response format
- SMTP over TLS (port 587) for email

7. Appendix

7.1 Live URLs

SERVICE	URL
Frontend	https://vtu-intern-2026-team15-java-fs.vercel.app
Backend	https://edutrack-backend-o1ym.onrender.com
GitHub	https://github.com/NaveenPR123/VTU_INTERN_2026_Team15_JavaFS